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Differences Between Stomata and Hydathodes

Stomata	Hydathodes
Occur in epidermis of leaves, young stems.	Occur at the tip or margin of leaves that are grown in moist shady place.
Stomatal aperture is guarded by two guard cells.	Aperture of hydathodes are surrounded by a ring of cuticularized cells.
The two guard cells are generally surrounded by subsidiary cell.	Subsidiary cells are absent.
Opening and closing of the stomatal aperture is regulated by guard cells.	Hydathode pores remain always open.
These are involved in transpiration and exchange of gases.	These are involved in guttation.



Can mangroove trees grow in salt water?

These amazing trees and shrubs cope with salt. Salt water can kill Plants, so mangroves must extract fresh water from the sea water that surrounds them. Many mangrove species survive by filtering out as much as 90 percent of the salt found in seawater as it enters their roots.

Mangrove excrete salt through glands in their leaves.

Halophiles

- Plants that grow in salty environment are called **Halophiles**.
- Plant growth in **saline habitat** developed numerous adaptations to **salt stress**. The secretion of ions by salt glands is the best known mechanism for regulating the salt content of plant shoots.
- Salt glands typically are found in **halophytes**. (Plants that grow in saline environments)



Figure 9.26: Halophytes



Figure 9.27: Removes excess salts through special salt glands on leaves